7 Proportions and percentage change

Sheet 39 Tue 2 Feb. 2027 *Proportions and percentages; proportion of a proportion*

1. Warm-up · Quick [PRO] Calculate 15 % of 240.
   Ans. $0.15 \times 240 = 36$.
2. Core · Method [PRO] 30 % of the pupils are boarders and 40 % of the boarders do dance. What proportion of the pupils are boarders who do dance?
   Ans. $0.3 \times 0.4 = 0.12 = 12\,\%$: proportion of a proportion (we multiply).
3. Core + · Problem [PRO] A school has 800 pupils, of whom 55 % are girls; 20 % of the girls study Latin. How many girls study Latin?
   Ans. $800 \times 0.55 = 440$ girls; $440 \times 0.20 = 88$.
4. Review · Quick [QP] 3 exercise books cost 4.50 €. Price of 7 books?
   Ans. $1.50 \times 7 = 10.50$ €.
5. Thread · True/false [EQ] True or false? Justify. The number -2 is a solution of the equation $3x + 1 = -5$.
   Ans. True. $3 \times (-2) + 1 = -6 + 1 = -5$.

Sheet 40 Thu 4 Feb. 2027 *Absolute change, multiplier, rate of change*

1. Warm-up · Quick [PRO] A price goes from 80 Ar to 92 Ar. What is the absolute change?
   Ans. $92 - 80 = 12$ Ar.
2. Core · Method [PRO] For the same change, give the rate of change and the multiplier.
   Ans. $t = \dfrac{12}{80} = 0.15 = 15\,\%$; $CM = 1 + t = 1.15$.
3. Core + · MCQ [PRO] A decrease of 20 % corresponds to the multiplier:
   A. 0.2 **B.** -0.8 **C.** 1.2 **D.** 0.8
   Ans. D. 0.8
   $CM = 1 + t = 1 - 0.20$.
4. Review · Quick [STA] Temperatures -2, 0, 3, 5, 9 (in °C). Mean and range?
   Ans. Mean 3 °C; range $9 - (-2) = 11$ °C.
5. Thread · Quick [SOL] Calculate the volume and the total surface area of a cube with edge 3 cm.
   Ans. $V = 3^3 = 27$ cm^3; $A = 6 \times 3^2 = 54$ cm^2.

Sheet 41 Mon 8 Feb. 2027 *Successive percentage changes*

1. Warm-up · Quick [CAL] Calculate 1.2×0.9.
   Ans. 1.08.
2. Core · Method [PRO] An increase of 20 % is followed by a decrease of 10 %. What is the overall rate of change?
   Ans. $1.2 \times 0.9 = 1.08$: overall increase of 8 %.
   Rates are not added; it is the multipliers that are multiplied.
3. Core + · True/false [PRO] True or false? Justify. An increase of 50 % followed by a decrease of 50 % returns to the initial value.
   Ans. False. $1.5 \times 0.5 = 0.75$: overall decrease of 25 %.
4. Review · Quick [SUB] Calculate $x^2 - x$ for $x = \dfrac{1}{2}$.
   Ans. $-\dfrac{1}{4}$.
5. Thread · Write-up [DEM] $ABCD$ is a parallelogram such that $AC = BD$. What type of quadrilateral is

it? Justify in one sentence.

Ans. Rectangle: a parallelogram whose diagonals have equal length is a rectangle.

Sheet 42 Tue 9 Feb. 2027 *Reverse percentage change; proportion and change*

1. Warm-up · Quick [CAL] Calculate $\dfrac{1}{1.25}$.

Ans. 0.8.

2. Core · Method [PRO] After an increase of 25 %, what decrease brings back the initial value?

Ans. $CM_{\text{rev}} = \dfrac{1}{1.25} = 0.8$: decrease of 20 % (not 25 %).

Pitfall: believing that an increase of 25 % is cancelled by a decrease of 25 %.

3. Core + · Interpret [PRO] Out of 250 candidates, 150 are successful; last year there were 200 candidates. Calculate the proportion of successful candidates, then the rate of change of the number of candidates.

Ans. Proportion $\dfrac{150}{250} = 60\,\%$ (part over whole); change $\dfrac{250-200}{200} = 25\,\%$ (relative change).

4. Review · Table [IMG] Table of values of f:

x	0	1	2	3
$f(x)$	4	1	0	1

Image of 2? Preimages of 1?

Ans. $f(2) = 0$; preimages of 1: 1 and 3.

5. Thread · MCQ [REP] On the globe, two points on the same parallel have:

A. the same latitude **B.** the same longitude **C.** the same distance from Greenwich **D.** the same altitude

Ans. A. the same latitude

Latitude: north/south angle from the equator; longitude: east/west angle from the Greenwich meridian.